

# API Gateway Benchmark Report

## Setup

Comparison of the current gateway build against the vendor's new release candidate under the enterprise launch traffic model (see load test results). Hardware-matched pods, production-mirror config, k6 generators in three regions.

## Results

| Metric          | Current   | RC        | Delta |
|-----------------|-----------|-----------|-------|
| p50 latency     | 34 ms     | 29 ms     | -15%  |
| p99 latency     | 210 ms    | 168 ms    | -20%  |
| Max RPS at SLO  | 48k       | 61k       | +27%  |
| CPU per 10k RPS | 3.1 cores | 2.4 cores | -23%  |

## Notes

Rate limiting accuracy improved under burst (no over-admission at the window edge). JWT validation offload reduces auth service load by 9%. Recommendation: adopt the RC behind a canary deployment in week 29, with rollback strategies pre-staged.